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MILITARY SPECIFICATION

MOUNT, TELESCOPE, M3A1; PACKAGING OF

1. SCOPE

1.1 This specification covers the packaging and packing of Telescope Mount M3A1, with or without Instrument Light M13 and Cover M410, for each level of protection.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

UU-T-106	Tape, Masking, Paper (Pressure-Sensitive)
UU-T-111	Tape, Paper Gummed (Sealing and Securing)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes; Wood, Cleated-Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock Corner
PPP-B-636	Boxes; Fiber
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant

FSC 1240

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Military

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-B-121	Barrier Material, Greaseproofed, Flexible (Waterproofed)
MIL-L-3150	Lubricating Oil, Preservative, Medium
MIL-G-10924	Grease, Automotive and Artillery

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and methods. - The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

3.1.1 Materials. - The following shall apply wherever packaging materials are referred to in this specification by name only:

REFERENCE	FULL IDENTIFICATION
Pressure-sensitive tape	Type I or II of Specification UU-T-106

3.2 Size of containers. - Bags, interior containers and exterior containers shall be of a minimum size that will adequately contain the items.

3.3 Preservation and packaging.

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3.3.1 Level A.

3.3.1.1 Cleaning. - Cleaning process C-1 of Specification MIL-P-116 shall be followed to clean the exterior surfaces of the mount, instrument light, and cover.

3.3.1.2 Drying. - Immediately after cleaning, the items shall be dried in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the items.

3.3.1.3 Preservative application. - Preservative conforming to Specification MIL-G-10924 (P-11 of Specification MIL-P-116) shall be applied on the unpainted surfaces of the mount. The preservative shall be applied by the brushing procedure specified in Specification MIL-P-116. Excess preservative shall be removed.

3.3.1.4 Unit packaging.

3.3.1.4.1 Mount with instrument light and cover. - Packaging the mount and cover shall be as specified herein, which is in conformance with Method IC-5 of Specification MIL-P-116. The instrument light shall be packaged as specified herein, which is in conformance with Method IA-8 of Specification MIL-P-116.

3.3.1.4.1.1 Covering preservative coated surfaces. - The preservative coated surfaces of the mount shall be covered or wrapped with greaseproofed barrier material conforming to type 1, grade A, class 2 of Specification MIL-B-121, which shall be secured in place with pressure-sensitive tape. The wrapped surfaces shall be overwrapped with greaseproofed barrier material conforming to type I or II, grade C, class 1 of Specification MIL-B-121. The wrap shall be secured in place with pressure-sensitive tape.

3.3.1.4.1.2 Level vial cover. - The level vial cover shall be rotated so that the vial is covered. The level vial cover shall then be secured in place with pressure-sensitive tape.

3.3.1.4.1.3 Instrument light. - The wire leads shall be coiled and secured to the instrument light body with pressure-sensitive tape. The instrument light shall then be wrapped with paperboard cushioning conforming

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to type I, class 1 of Specification PPP-P-291. The wrap shall be secured in place with gummed tape conforming to class 2 of Specification UU-T-111. The wrapped light shall then be placed in a water-vaporproof bag conforming to type II, class e of Specification MIL-B-117. The bag shall be heat sealed in such a manner as to pass the heat seal and quick leak tests of Specification MIL-P-116. Entrapped air shall be exhausted prior to final closure.

3.3.1.4.1.4 Cover.- The cover shall be folded in such a manner that it will fit the void shown in figure 3.

3.3.1.4.1.5 Immobilizing the mount, instrument light and cover in a fiberboard box.- The fiberboard box in which the items are placed shall conform to class 2 or 3, any compliance symbol, and any style, except FTC, TS, OPF and FPF, of Specification PPP-B-636. The size of the box shall be as specified in figure 1. The items shall be immobilized in the fiberboard box by utilizing a fiberboard pad and die cuts as specified in figure 1. The pad and die cuts shall be fabricated from corrugated fiberboard conforming to class 2 or 3, of Specification PPP-B-636. The pad, items, and die cuts shall be positioned in the fiberboard box as shown in figure 3. The fiberboard box shall be closed with pressure-sensitive tape conforming to Specification PPP-T-76 in accordance with the appendix of Specification PPP-B-636.

3.3.1.4.2 Mount without instrument light and cover.- Packaging the mount without the instrument light and cover shall be as specified in 3.3.1.4.1.1, 3.3.1.4.1.2 and 3.3.1.4.1.5, except that references to packaging the instrument light and cover in 3.3.1.4.1.5 shall be omitted.

3.3.2 Level C.

3.3.2.1 Cleaning, drying, and preservative application.- Cleaning, drying, and preservative application shall be as specified in 3.3.1.1, 3.3.1.2 and 3.3.1.3, except that preservative conforming to Specification MIL-L-3150 (P-7 of Specification MIL-P-116) shall be used in lieu of the preservative specified in 3.3.1.3.

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3.3.2.2 Unit packaging.

3.3.2.2.1 Mount with instrument light and cover. - Packaging shall be as specified in 3.3.1.4.1.1 to 3.3.1.4.1.5 inclusive, except as follows, which is in conformance with Method I of Specification MIL-P-116 for the mount and Method III of Specification MIL-P-116 for the instrument light and cover:

(a) The pressure-sensitive tape, specified in 3.3.1.4.1.2, shall be omitted.

(b) The water-vaporproof bag, specified in 3.3.1.4.1.3, shall be omitted.

(c) A fiberboard box conforming to type I or II, class 1, any style of Specification PPP-B-636 shall be used in lieu of the box specified in 3.3.1.4.1.5. The pad and die-cuts shall be fabricated from corrugated fiberboard conforming to type I, class 1 of Specification PPP-B-636. The fiberboard box shall be closed as specified in the appendix of Specification PPP-B-636.

3.3.2.3 Mount without instrument light and cover. - The mount without the instrument light and cover shall be packaged as specified in 3.3.2.2.1, except that references to packaging the instrument light and cover shall be omitted.

3.4 Packing.

3.4.1 Level A. - Twelve packaged mounts, with or without the packaged instrument lights and covers, shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-601 - Style A or B (oversea type) - type II or III, class 2 plywood
PPP-B-621 - Class 2 (oversea), style 2 or 4

3.4.2 Level B. - The requirements of 3.4.1 shall apply, except that the shipping container shall conform to one of the following specifications:

PPP-B-585 - Any style, except style 1, class 1, construct for type 3 load
PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic), style 2 or 4

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3.4.3 Level C. - A number of unit packaged mounts, with or without the packaged instrument lights and covers, shall be immobilized in a fiberboard shipping container conforming to type I or II, class 1, any style, except FTC, TS, OPF and FPF of Specification PPP-B-636. The gross weight shall not exceed the weight limits specified in Specification PPP-B-636. When the quantity of mounts to be shipped is less than twelve, the fiberboard unit container can be used as a shipping container.

3.4.4 Strapping. - Strapping shall be applied on shipping containers. The type of strapping and the method of applying it shall be as specified in the applicable container specification.

3.5 Marking.

3.5.1 Interior packages. - Marking each bag and each fiberboard box shall be in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A unit package and omitted on the Level C unit package.

3.5.2 Shipping containers. - Each shipping container shall be marked in accordance with Standard MIL-STD-129. If the serial numbers of the mounts in a Level A or B shipping container are in consecutive order, the serial numbers shall be indicated as specified in Standard MIL-STD-129. If the serial numbers are not in consecutive order, they shall be indicated on a packing list.

3.5.3 Stock number. - The stock number or numbers to be added in the identification data shall be obtained from the contracting officer.

3.6 Preproduction pack. - Prior to quantity production, a preproduction pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification. A rough handling test is required.

3.7 Workmanship. - All operations involved in accomplishing corrosion prevention, preservation, packaging, and packing by the procedures specified herein shall be in accordance with the highest grades of practice associated with this type of work.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Contractor's quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

(e) Measures for preventing any significant departure of product quality from standards.

4.1.1.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance.

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Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Control of materials. - Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.3 Inspection provisions. - Quality assurance provisions as specified in Specification MIL-P-116 shall apply where applicable.

4.3.1 Lot formation. - Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.2 Acceptance tests. - The classification of defects in tables I and II shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.3.2.1 Disposition of rejected lots. - Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE I - CLASSIFICATION OF DEFECTS

For Normal Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105
For Reduced Inspection Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105
AQL values are expressed in percent defective

MAJOR: AQL 1.5%	Req't Par.	Test Method
1. Improper cleaning (Level A, C)	3.3.1.1 3.3.2.1	MIL-P-116 Visual
2. Improper drying (Level A, C)	3.3.1.2 3.3.2.1	MIL-P-116 Visual
3. Improper preservative application, not conforming to specification, excess preservative not removed (Level A, C)	3.3.1.3 3.3.2.1	MIL-P-116 Visual
4. Unit packaging, mount and cover incorrect method (Level A)	3.3.1.4.1	MIL-P-116 Visual
5. Unit packaging, instrument light, incorrect method (Level A)	3.3.1.4.1	MIL-P-116 Visual
6. Preservative coated surfaces inadequately wrapped, not as specified, barrier material not conforming to specifications, not secured (Level A, C)	3.3.1.4.1.1 3.3.1.4.2 3.3.2.2.1 3.3.2.3	Visual
7. Vial not covered and secured as specified (Level A, C)	3.3.1.4.1.2 3.3.1.4.2 3.3.2.2.1 3.3.2.3	Visual
8. Instrument light, wire leads not coiled and secured as specified, paperboard cushioning not conforming to specification, wrap not secured, tape not conforming to specification (Level A, C)	3.3.1.4.1.3 3.3.2.2.1	Visual

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TABLE I - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 1.5%	Req't Par.	Test Method
9. Instrument light, water-vaporproof bag not conforming to specifications, unsealed, punctured, improperly sealed, excess air not expelled (Level A)	3.3.1.4.1.3	MIL-P-116 Visual
10. Mount, instrument light and cover not immobilized as specified, fiberboard box not conforming to specification, box size not as specified, pad and die cuts not fabricated as specified, pad, items and die cuts incorrectly positioned, not as illustrated, box closure not as specified (Level A, C)	3.3.1.4.1.5 3.3.1.4.2 3.3.2.2.1 3.3.2.3 Figure 1 Figure 3	PPP-B-636 Visual
11. Unit packaging; mount, incorrect method (Level C)	3.3.2.2.1 3.3.2.3	MIL-P-116 Visual
12. Unit packaging; instrument light and cover, incorrect method (Level C)	3.3.2.2.1	MIL-P-116 Visual
13. Packaged mounts with or without packaged instrument lights and covers not immobilized, shipping container not conforming to specification, incorrect quantity (Level A, B, C)	3.4.1 3.4.2 3.4.3	PPP-B-585 PPP-B-601 PPP-B-621 PPP-B-636 Visual
14. Unit packaged mounts with or without packaged instrument lights and covers, excess weight, incorrect quantity and container (Level C)	3.4.3	PPP-B-636 Standard Measuring Equipment (SME) Visual
15. Shipping container not marked in accordance with standard, serial numbers not as specified, omitted, incorrect, illegible (Level A, B, C)	3.5.2	MIL-STD-129

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TABLE I - CLASSIFICATION OF DEFECTS (Cont'd)

MINOR: AQL 4.0%	Req't Par.	Test Method
41. Cover not folded as illustrated (Level A, C)	3.3.1.4.1.4 3.3.2.2.1 Figure 3	Visual
42. Strapping; incorrect application and method (Level A, B, C)	3.4.4	Visual
43. Marking; interior packages not in accordance with standard (Level A, B, C)	3.5.1	MIL-STD-129 Visual
44. Serial number omitted, incorrect, illegible (Level A)	3.5.1	Visual
45. Stock number omitted, incorrect, illegible (Level A, B, C)	3.5.3	Visual
46. Workmanship	3.7	Visual

TABLE II - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level L6, and Sampling Plan Table IV-A of Standard MIL-STD-105
AQL value is expressed in percent defective

MAJOR: AQL 4.0%	Method of Unit Protection	Test Method
Heat seal	IA-8	MIL-P-116
Quick leak	IA-8	MIL-P-116

4.3.3 Rough handling tests .- The preproduction pack (specified in 3.6) shall be subjected to the "Rough Handling Tests" in accordance with Specification MIL-P-116, applicable only to level "A".

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4.3.3.1 Preproduction pack. - The preproduction pack (specified in 3.6) shall be inspected 100 percent by the contractor or his representative, and witnessed by the Government inspector to determine conformance with the requirements of this specification for Level A.

4.3.3.2 Preproduction pack failure. - Should the preproduction pack fail to pass the test for Government acceptance, packaging operations will cease until the defective characteristics have been corrected in the process used to produce the preproduction pack. When corrections in the process have been made, a new preproduction pack shall be completely inspected by the contractor and witnessed by the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 There are no applicable requirements.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of packaging and packing

6.2 Dimensions, cubage and gross weight of shipping container.

6.2.1 Mount with instrument light and cover. - The following inside dimensions, overall cubage, and gross weight are approximate for twelve unit packages packed in a nailed wood box for Level A:

Inside dimensions - 25-1/2 inches long by 24-1/2 inches wide by 13-1/4 inches deep
Cubage - 6.2 cubic feet
Gross weight - 200. pounds

6.2.2 Mount without instrument light and cover. - The following inside dimensions, overall cubage, and gross weight are approximate for twelve unit packages packed in a nailed wood box for Level A:

Inside dimensions - 25-1/2 inches long by 24-1/2 inches wide by 13-1/4 inches deep
Cubage - 6.2 cubic feet
Gross weight - 185. pounds

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6.2.3 Definition.

6.2.3.1 Standard measuring equipment (SME). - Standard measuring equipment (SME) includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

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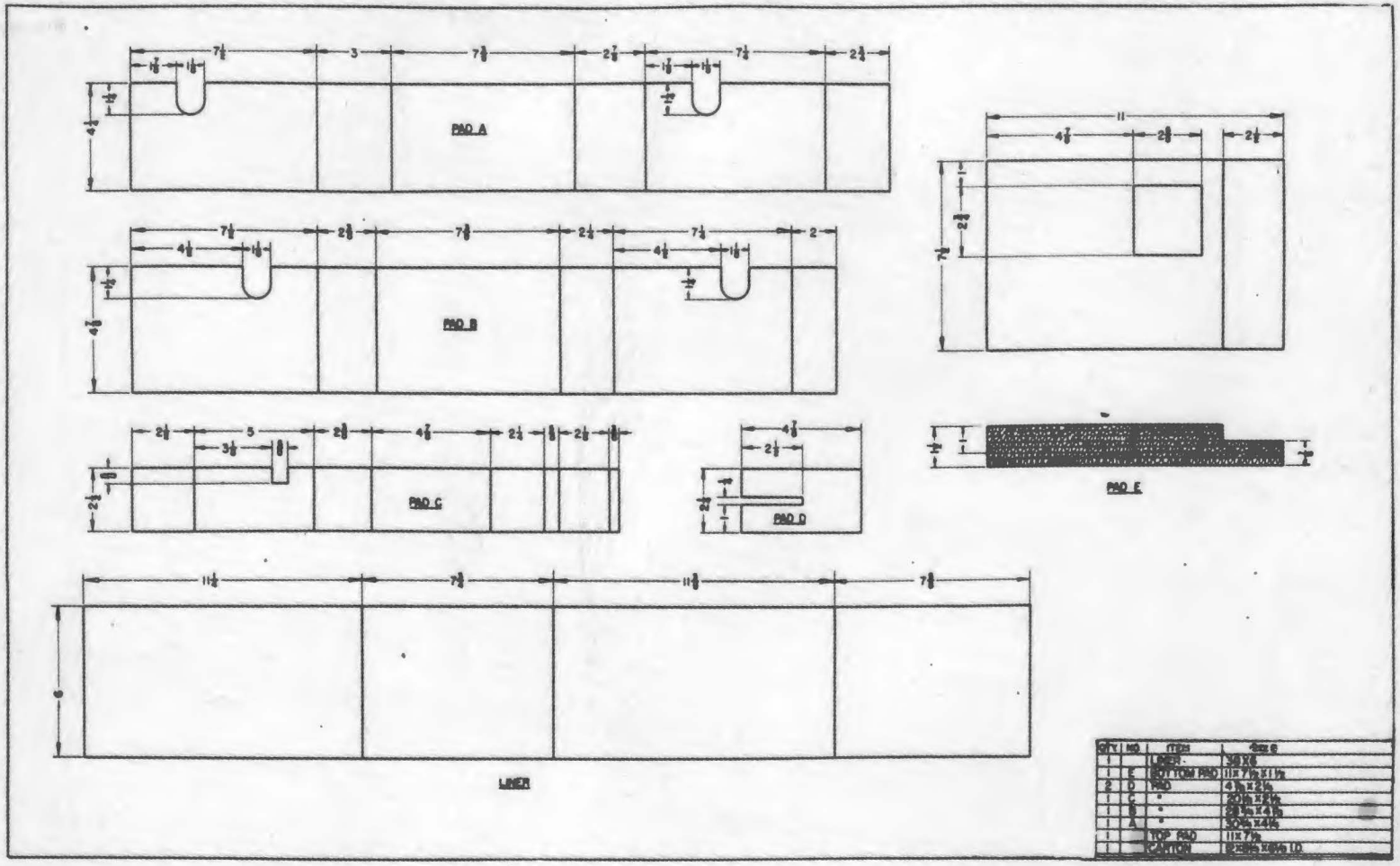
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36.231.S2521/ORD.59

CORRUGATED PAD and DIE CUTS

FIGURE 1

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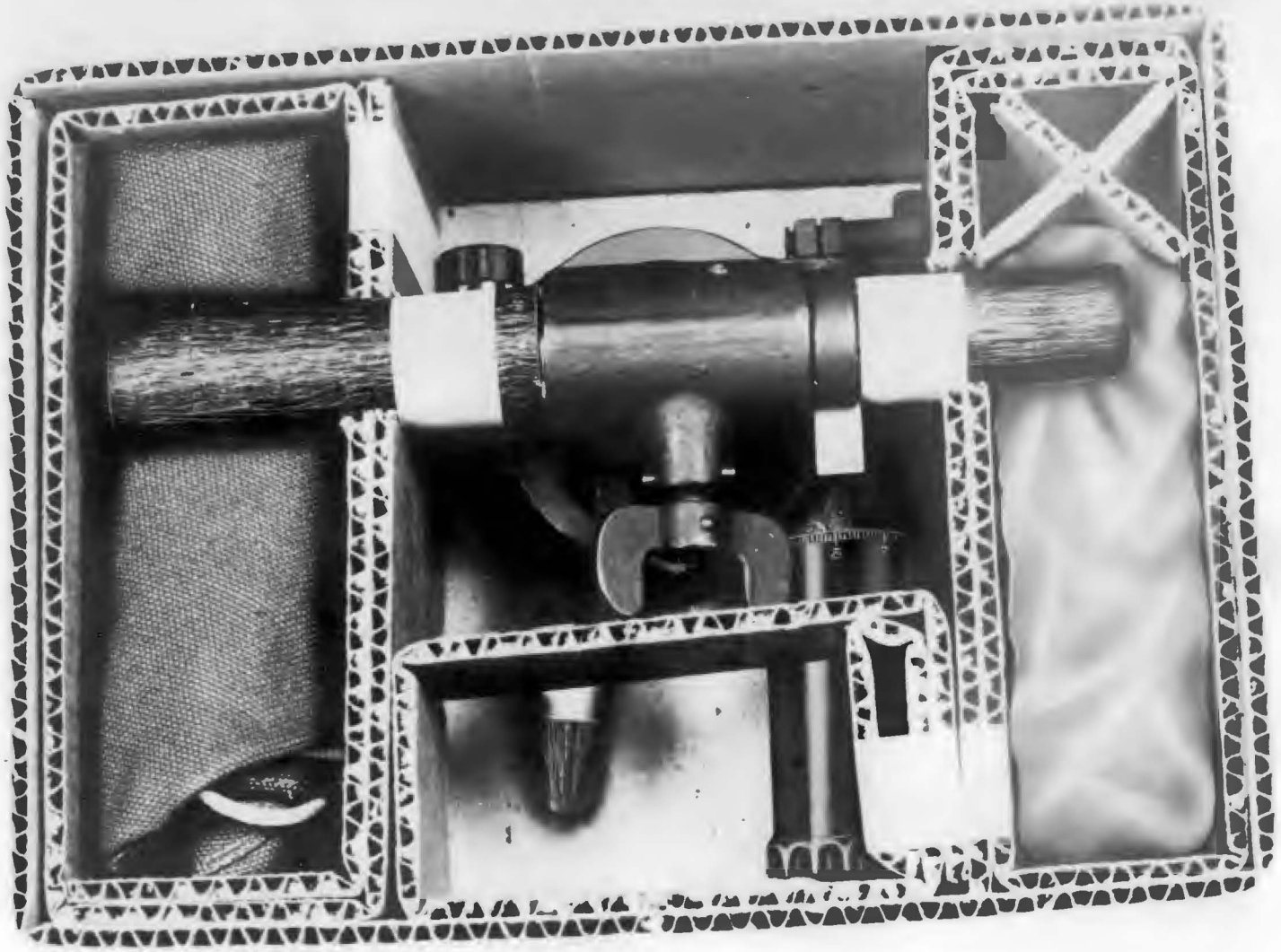
36.231.S2519/ORD.59

MOUNT, TELESCOPE, M3A1 - Telescope mount with instrument light and cover.

FIGURE 2

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MIL-I-6516 (ORD)



36.231.S2520/ORD.59

MOUNT, TELESCOPE, M3A1 - Positioning of items; corrugated die-cuts and Pad as they are placed in fiberboard box.

FIGURE 3

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